Project 3: Analysis of Instant Messaging Apps

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Data Collection

Brainstorming

For this project, our team wanted to focus on applications, rather than objects. Each of us suggested different programs that we use on a relatively regular basis, then grouped them together under similar umbrella terms to form a **mind map** (*Figure 1*). We wanted to simplify our interview process by selecting applications used by the average student, such as Google Docs, Instagram, the Weather app, and Spotify. While these were good options, we decided to focus on apps that had messaging functionalities, allowing for students to communicate with each other virtually. Not only do a majority of students interact with communication tools on a frequent basis, it has an array of visual, tactile, and auditory features that we can analyze from a design perspective. Since **affordances** can never be truly universal, we are hoping to glean what **errors** arise when trying to design an app that functions for the most diverse amount of users as possible.

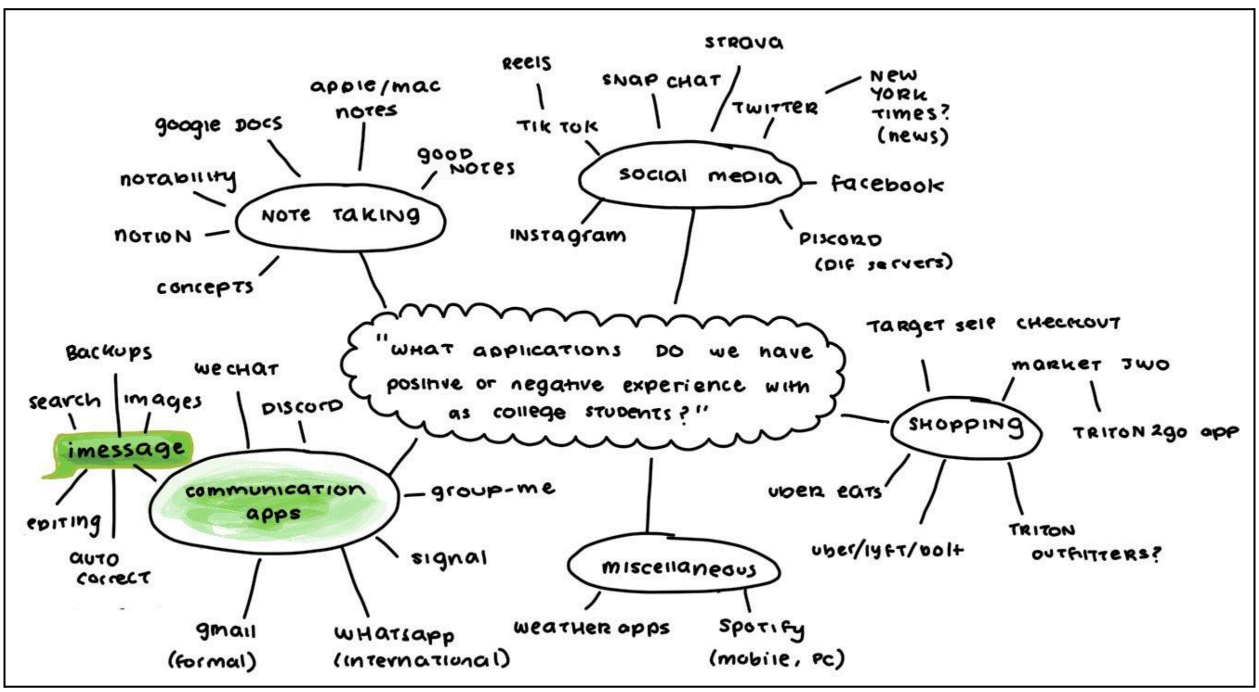

Figure 1. Mind map considering various applications

Methodology:

To begin our project, we started brainstorming potential pre and post-interview questions and finalized our four tasks. We decided to not focus our analysis on one application, iMessage, but expanded our observations to text based communication in general, including platforms like Discord, WhatsApp, Facebook Messenger and more. From there, we began creating a list of non-leading questions needed during the interview process. We identified some features and functionalities of chat apps that our project members have had **errors** with in the past such as customizing built in features, managing conversations, and sending and receiving attachments. We wanted to interview a diverse array of people. Age was the first important demographic factor. We were interested in how people of different ages, genders, and backgrounds have different **user experiences** with modern communication technology. We didn't know if virtual interviews would affect how users respond to instructions to use an already digital platform. Thus, we conducted interviews both online and in person, so we could have a balanced data set.

Pre-Task Questions:

Our first set of pre-task questions asked participants if they experienced any issues with unsending, autofill, and pretty general introductory questions more specific to communications across platforms. We want to gauge their preexisting **mental models** of messaging.

1. Have you ever messaged the wrong person? Elaborate, what app?
2. Do you rely on autofill? Does autofill make you stumble?
3. Do you use built-in keyword search engines? Vs scrolling through the chat. Elaborate.
4. Do you have any way to backup chat communications? Have you deleted a message and needed it later?
5. Have you tried to delete a message and it failed? How did it go?

iMessage Tasks:

Now that we have an idea of **errors** they have experienced in the past, we can compare it to new **errors** encountered in our tasks. We focused on how participants interact with their devices. To gather an understanding about how attachments are sent, we had them send us an image and documented the process. Our second task had users typing in specific keywords and phrases that are prone to autocorrect. We chose this task because the specific keywords trigger Apple's autofill feature and make it easier for observers to note difficulties under a controlled setting. Our last two experiments tested users about their knowledge of the search function and the undo feature. We wanted to see if users ran into difficulties undoing their actions or searching for specific dates due to **slips**, **knowledge-based mistakes** (deeper misunderstanding or lack of experience), or **constraints** of the product

1. Send any attachment through the platform of choice
   a. Example: 10th image in photos
2. Rewrite these words of phrases
   a. Thanos > Thanks
   b. Omw > "On my Way!"
   c. I have a Problen with tis spellinb > I have a problem with this spelling
3. Write out a sentence, then delete part of it or the whole thing
   a. How would you go about undoing your deletion?

4. Keyword and Date search
    a. Show us a message with "Birthday" in it
    b. Find a text you sent on Wednesday, October 30th
    c. We timed both of these tasks and took note of user demographics

Post Task Questions:
Our post-test questions wrapped up key points and expanded upon users' understanding of and experience with other messaging platforms. Getting an idea of the **feedback** used by participants throughout the interview process helped us determine how well they crossed the **Gulf of Evaluation**.

1. What did you think of these tasks?
2. Do you use cross platform communication with the same person? Do you ever juggle conversations?
3. Please select your top 3 apps from this list, and rate them 1-5 below. 1 being the lowest, and 5 being the highest.
    a. iMessage
    b. Discord
    c. WhatsApp
    d. Zoom Chat
    e. Gmail
    f. ChatGPT
    g. Instagram Direct Messaging
    h. Facebook Messenger

Our interviewees, of diverse backgrounds and generations, are all Apple users. We conducted interviews through in-person and virtual methods, such as phone calls or Zoom meetings. We utilized a Google form and subsequent sheet to document our observations and keep track of each question and task. Participants were all asked the same questions and were given appropriate time to formulate answers. To remain unbiased and to avoid swaying interviewee responses, interviewers took the role of the apprentice in the **master-apprentice model**, listening to respondents as if they were learning about the process through their verbal observations.

Proof of data:

A link to our google sheet can be found here. Our data collection sheet and evidence of each member completing at least three individual interviews can be found here. Our graphs and numerical summaries can be found here.

Contributions

Amelia refined and expanded upon the methodology section, compiled proof of data, and wrote the class concept analysis. Elana worked on Brainstorming, Trade-offs, Quantitative data graphs, and the Patterns/Trends section, as well as overall clarity and format. Alex created the google form to capture our data, some graphs used in the Patterns/Trends section, collected images for Error Analysis, and created the Design Space Chart alongside its corresponding tables and explanations. Matt created the interview questions and tasks, double checking accuracy with the TAs and writing hub in subsequent meetings during Week 5. Matt started writing the methodology section, Redesigns, and **trade-offs**. Matt went to the writing hub meeting 11/10/2024 and office hours 11/12/2024.

Error Analysis

Identify/classify Errors:

Through our task observations and interview questions, we gathered 5 major errors. **Slips** and **mistakes** usually arise due to a complex and/or confusing **system image** in which users are unsure how to cross the **Gulfs of Execution and Evaluation** to interact with a product because the interface is not intuitive. With digital interfaces especially, most users have a preexisting **mental model** that predicts how a system works, which only makes usage more complex when features are added and removed. Additionally, incorporating new features and **signifying** their existence to users becomes increasingly more difficult with the device's limited real estate. Norman calls this the **Paradox of Technology** because these features are supposed to make life easier but can end up causing more confusion.

Figure 2. A screenshot displaying the lack of dismissive options regarding autocorrect. Here, "Omw" is automatically suggested as "On my way!" However, users are forced to either accept it by pressing space or send, or reject it by manually backspacing or clicking the original input on the left.

Figure 3. In this instance the task revealed the spell checker **error** when talking about the character Thanos. The user intended to write "Thanos" and was forced to write "thanks"

For the first task, participants typed out intentional phrases or words that would trigger Apple's autocorrect feature such as misspellings. When asked to type "Thanos", which autocorrects to "Thanks", users accidentally sent the wrong word. This resulted in an **action-based slip** as an unintended action was performed. Similarly, when asked to override Apple's autoreplacement feature for the acronym "omw", users repeatedly meant to send "omw" instead of the autocorrected "On my Way!" (see: *Figure 3*). One user had repeated issues with the autocorrect tasks and tried to type slower to catch the autofill answers before they replaced her text, but even after multiple attempts, the app kept automatically autocorrecting her inputs. Because they intended to send one thing but instead triggered another response accidentally, this would also be considered a **memory lapse slip**. Although the software affords rejecting – *backspaces, suggested words tab* – and accepting autocorrect suggestions – *space bar, send button* – there is no clear **signifier** that opts to ignore the suggestion (see: *Figure 2*). The **Gulf of Execution** is not bridged because, while users have clear intent going into sending their message, they are unable to perform the action sequence as intended.

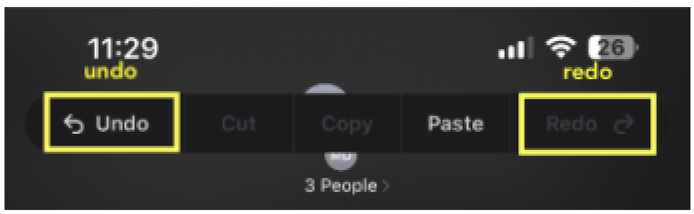

Figure 4. A screenshot displaying the menu that appears with a three-finger tap, offering options like Undo and Redo.

In another task, users were asked to type out a sentence and accidentally delete a part of it. We observed that 9/16 users did not know the shake to undo feature, which is Apple's feature that undoes edits and is triggered when a user shakes their phone. Additionally, 1/16 users knew that a three-finger tap triggered another menu at the top of their screen with an undo and redo button (see: *Figure 4*). Only about 6/16 users utilized the shake to undo feature. Participant 1 was surprised about the shake feature, remarking that they had seen the feature appear on their phone on accident but did not know that function was an undo feature. Because the users had an incomplete understanding of the undo shortcut options, this error would be considered a **knowledge-based mistake**. This mistake is mostly due to invisible **signifiers**; iMessage **affords** multiple ways to undo edits but does not clearly **signify** where users can access this tool. The existence of multiple settings menus with unclear **signifiers** may be solved by synthesizing them into a single menu, such as the hover menu that appears upon a single tap (*Figure 4*). The three finger tap and phone shake are either **discoverable** accidentally, or not obvious enough to be discovered by a majority of our participants at all.

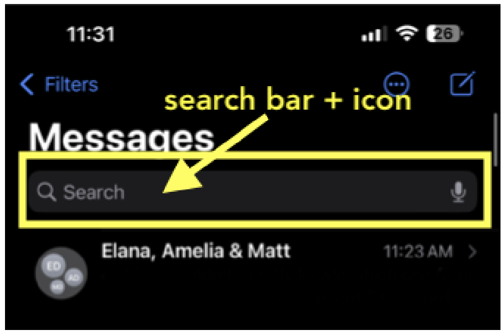

Figure 5. A screenshot illustrating the search function in iMessage. The magnifying glass icon and "Search" text are only visible when a user scrolls up and are sometimes not visible upon opening the app.

In the final task, users timed when asked to locate a message with the word "birthday" in it. They were then asked to locate a message from October 30, 2024. 12/16 participants did not have any issues with the "birthday" task but for the second task, 6/16 successfully utilized the search function and 10/16 scrolled up manually. For those manually scrolling, scroll times averaged 15.42 seconds. Participant 5, who took over a minute for both tasks, remarked during the wrap up questions that these two tasks were especially challenging for her. This lack of familiarity with the search tool, causing people to scroll through dozens of texts, is a **knowledge-based mistake** because users did not know that there is a dropdown search feature, causing them to revert to less efficient methods of message sorting. This mistake is due to the poor **discoverability** of the search function, which only appears on iMessage's home screen; the magnifying glass icon and "Search" text **signifier** are only visible when a user scrolls up and are not visible upon opening the app (see: *Figure 5*). While the app **affords** searching, the **spatial mapping** could be better placed so they are **discoverable** and intuitive to users.

Patterns/trends:

The most prevalent pattern we encountered was among users who messaged the wrong person. 14/16 interviewees admitted they had sent text, emails, or DMs to the incorrect recipient, to varying levels of severity and embarrassment. This is a trend that supersedes demographics such as age or gender, thus we can attribute it to faulty design. We attributed this pattern to the design's small interface, which makes it hard for a user to see who they are messaging, juggle multiple email addresses or phone numbers, and avoid **action slips**, etc. Considering the application has no way of predicting your intended recipient, the **feedback** to detect an erroneous message is minor. Users must rely on visual feedback and their own intuition to determine when this error has occurred. If they are distracted, the consequences can grow. Thus, this trend is difficult to resolve because there are no obvious design features that would make bridging the **Gulf of Evaluation** easier.

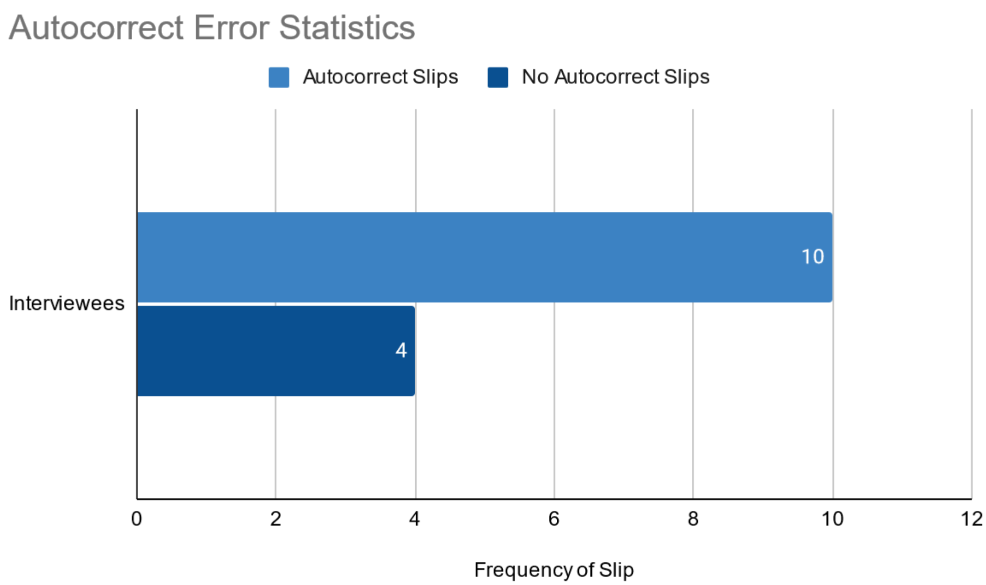

Figure 6. A bar chart showing the frequency of **slips** among interviewees. The majority of participants (12/16) experienced autocorrect-related errors, while a smaller portion (4/16) reported no autocorrect slips.

Furthermore, as shown in *Figure 6*, over half (11/16) of our users encountered difficulties with grammar correction. The prevalence of this pattern supports the notion that the auto correction algorithm is poorly designed. And yet, 12/16 users reported frequently relying on autofill. Even when it causes trouble, autocorrect is the preferred choice. In other words, autocorrection is a popular design feature for instant messaging apps, and tends to be worth the significant amount of **slips** it produces. User 15 mentioned that autocorrect often types a different, more commonly used word than the one they wanted (**capture slip**), or doesn't even know what outcome to give a very misspelled or foreign word (**rule-based mistake**).

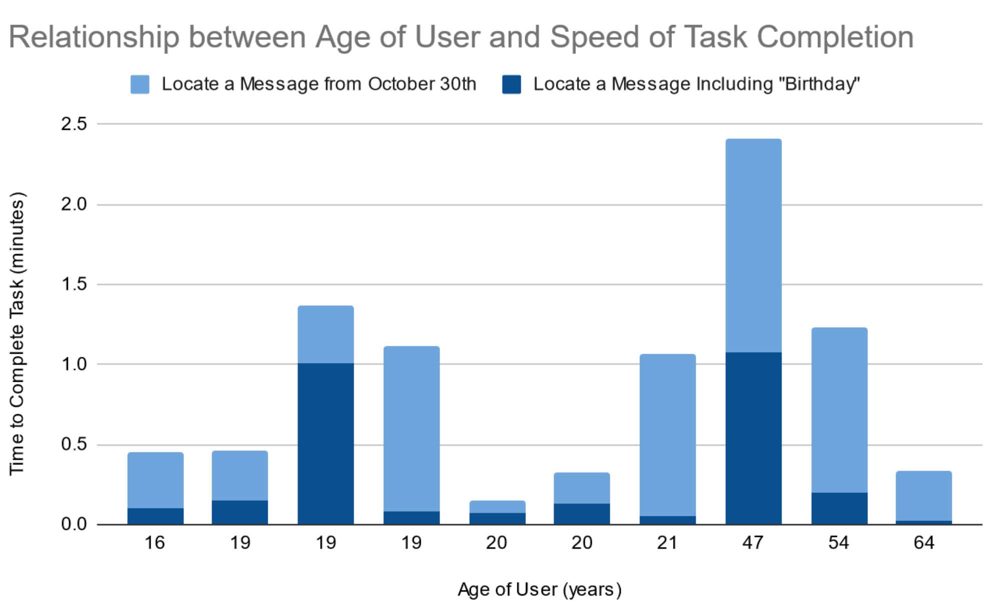

Figure 7. A bar chart displaying the distribution of minutes it took for interviewees to locate a message from October 30th and locate a message including the word “Birthday.”

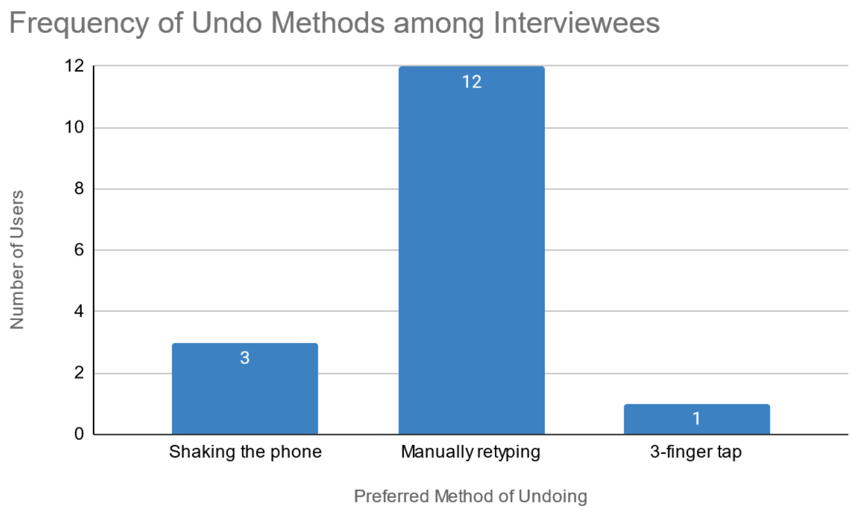

Figure 8. A bar chart displaying the distribution of undo methods among users. The majority (12/16) relied on backspacing to correct their sentence deletion, 3/12 used the shake-to-undo feature, and only one person utilized the three-finger tap menu to undo. This reveals the wide familiarity of methods for undoing deletion.

As discussed in the problems section, **knowledge mistakes** were prevalent among our interview responses. We wanted to further investigate whether there was a demographic pattern to this error: what is the relationship between the age of the participant and the amount of time it takes them to search a message? For instance, task 4 asked users to locate any message from approximately one week ago. This task tested the familiarity of the user with the search engine and message-location features. We theorized that older participants would have less understanding of the algorithm. Thus, when faced with unfamiliar **signifiers** and **affordances,** they wouldn’t be equipped to bridge the **Gulf of Execution**. We also took this trend further to investigate how familiar were interviewees with the undo feature; a similar knowledge based process.

Figure 7 shows a bimodal distribution of time for task completion, relatively balanced between older and younger users. This leads us to conclude that difficulty with search algorithms is a design flaw that transcends demographic differences. The designer’s **conceptual model** could be improved to make this feature more intuitive and efficient. *Figure 8* further illustrates this argument. A vast majority of participants were not aware of implemented undo methods. When these different methods have no meaning to a user, they become bound by **semantic constraints**. In other words, the user is blind to these functions when they are not properly communicated by the **system image**. This trend of poor signifiers is one that applies across patterns in instant messaging applications. As discussed in the previous section, this leads to **mistakes** where the user's **mental model** is faulty to begin with, and **slips** where their goal was correct but some design element or external circumstance disrupted them.

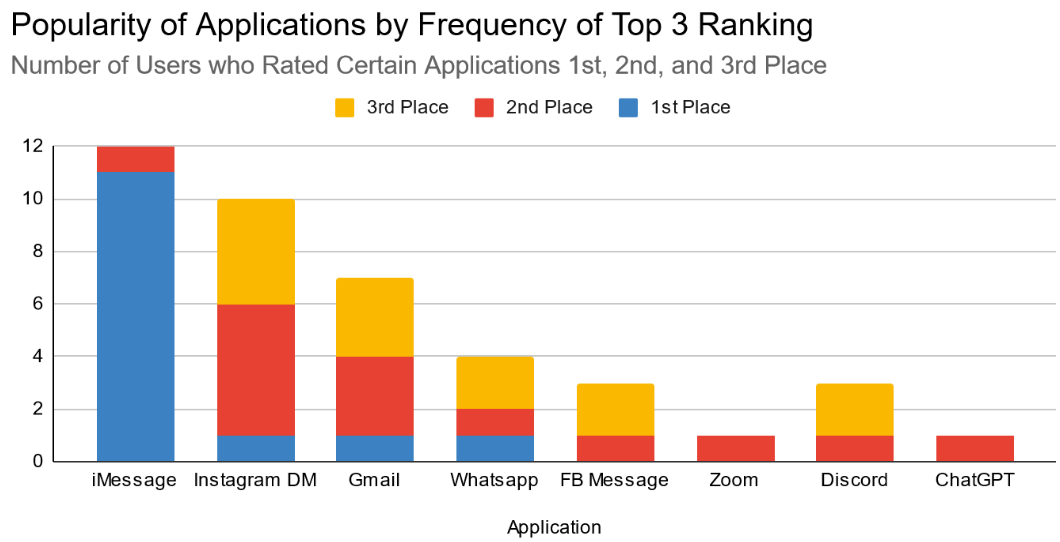

Figure 9: A bar graph depicting the user's top rated apps by how they ranked in a popularity poll

Overall, despite errors encountered with iMessage, 11/16 users rated it their number one app. A possible explanation for this pattern is the importance of familiarity in design usage. Apple has retained a characteristic interface and reused certain functions for many years. There are many **cultural constraints** around using certain instant messaging apps over others. People can be limited to designs used by those around them. Users who grew up relying on this application are more likely to tolerate when **slips** inevitably occur, as opposed to newer apps like Zoom and ChatGPT, who's novelty comes off as overcomplication. Instagram was the most popular app for second place. Perhaps this implies its nature as a supplementary app to iMessage. Users 1, 2, 3, 15 and 16 all reported that they used Instagram to supplement more serious conversations happening on iMessage. On social media, they could share and discuss Reels, posts, and general lighthearted content. Then, as conversations moved to iMessage, these users shifted towards more primary conversation. Gmail established a trend of being an important but not vital application in our interviewee's experience (see: *Figure 9*). It functioned as it needed to for professional contexts, but was less prevalent in their daily life than iMessage.

Tradeoffs:

The instant messaging industry is saturated with a diversity of designs, user interfaces, and operational functions. As a consequence, designers must juggle the drawbacks of increasing the quality of one attribute at the expense of another; also known as a **tradeoff**. We took note of the most prominent ones among our user's experiences.

<u>Hidden vs Obvious Operations</u>

Many of our users were surprised to learn about certain messaging functions, such as the three finger tap, shaking, etc. The designer's **conceptual model** included these operations in the device, but they are not obviously **signified** anywhere. In other words, they are "hidden" from the **system image**, unable to be perceived by the user. A benefit of hidden operations is that they give the user access to a wider range of functionality. With limited space and visibility, the designer must give priority to certain operations, like sending, over others, like undoing. These hidden techniques make the design more powerful, but according to *The Design of Everything Things* a signifier that cannot be discovered by the user is effectively useless (19). Hidden functions, while increasing the potential capability of a design, must be obvious enough to enable the user to bridge the **Gulf of Execution**. Otherwise, confusion over **signifiers** and **affordances** will prevent the user from developing the correct **mental model**. This could easily lead to **knowledge-mistakes**. On the other hand, making every function obvious will also clutter the design and confuse the user equally. This **tradeoff** is all about balancing functionality with communication.

Manual Correction vs Autocorrect

This is both a tradeoff in terms of **user experience** and **conceptual models.** Powerful autocorrect systems allow the user ease of mind. They have to put less individual effort into their messaging, allowing them to engross themselves in the conversation. Yet, these systems also take autonomy away from the user. As discussed in previous sections, oftentimes autocorrect acts against a user's will. User 5 and 14 both reported sending messages with erroneous autocorrects, and not noticing until it was too late. For example, changing "ruining the mood" to "running the mood" in User 14's experience. While these algorithms increase the speed and efficiency with which online communications take place, they can cause annoying or embarrassing errors. Manual control is more time consuming, but at least the user can regulate their own mistakes. It is ultimately the user's decision to choose the **tradeoff** they want: turning autocorrect On or Off. Yet, designer's must also be sympathetic of the drawbacks a user faces when it overcorrects. Their **conceptual model** might include a powerful automated system, while the user's **mental model** includes personalized chat.

Efficiency vs Accuracy

This is the broadest tradeoff we encountered, as it applies to messaging history, autocorrect, communicating with multiple people, etc. How do instant messaging apps balance the efficiency with which a task can be carried out against the accuracy of the results? For instance, task 4 asked users to locate any message from approximately 10 days ago. Users who scrolled through their message history found the exact text from exactly 10 days ago, but it was more time consuming. Users who looked it up with the search feature got there faster, but only through texts from around the same time. In other words, low accuracy. There is no design feature that integrates history searching with speed and precision. This is also a tradeoff between **knowledge in the world**, being the physical operational abilities of a design, and **knowledge in the head**, being a user's desire for efficiency and their **mental model** on how to achieve that.

Design Space & Redesign

Design Space Chart:

The designs that we have chosen to compare in our Design Space were iMessage, Instagram, and Discord. iMessage is our most popular application that was used for a majority of our interviewees' tasks. Discord is a popular chatting application that is used on the computer — a device that differentiates from a phone which is commonly used. Instagram DMs is also an interesting design because it is embedded into a social media app that is commonly used amongst college students. All together, these designs provide a diverse perspective of different messaging applications for our design space chart.

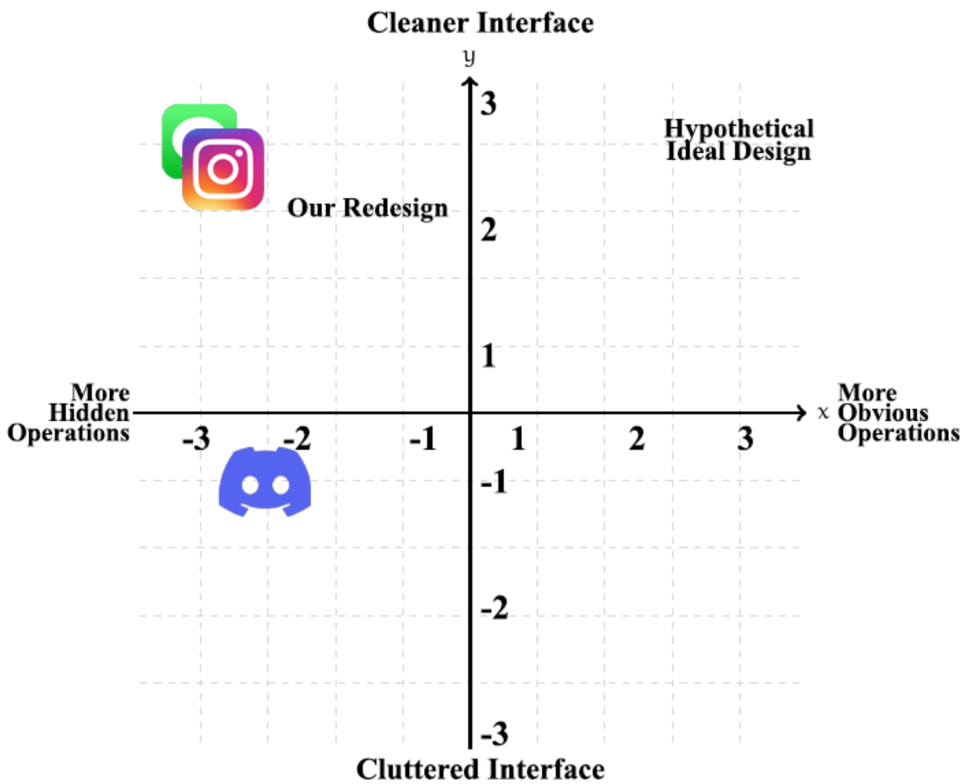

Figure 10. Our first design space chart that targets the **tradeoffs** between having hidden vs obvious operations, this coming at a cost of interface simplicity.

Chat Application	**Operation Indicators (X-Axis)**	**Interface Simplicity (Y-Axis)**	**Explanation**
iMessage	-3	3	iMessage has a -3 for operation indicators because it has a lot more hidden features. For instance, interviewee #4 had no clue about the shake to undo feature when it came to undoing a deleted portion of their message. Even more so there is the 3-tap gesture that reveals a menu with undo and redo buttons, which not everyone knows about because there is no clear **signifer** to activate it. It received a 3 for interface simplicity because it has a cleaner interface, there may be no consistent menu with undo or redo buttons, but at least there is a lot of space to simply read and send messages.

Discord	-2.5	-1	Discord has a -2.5 for operation indicators because it has less hidden operations than iMessage, although some operations like undo and redo could be visible for those who don't know keyboard shortcuts for them. It has a plethora of visible features that iMessage does not have. On the other hand, it received a -1 for interface simplicity because at the cost of having these less hidden features, these icons and buttons take up more of the screen. Although Discord is meant for computer users, this may not be as much of a problem.
Instagram	-3	3	Instagram's direct messaging design is quite similar to iMessage. So like iMessage, it received a -3 for operation indicators because it has more hidden features like lack of signifiers to undo or redo messages. It received a 3 for interface simplicity because it has a cleaner interface. As aforementioned with iMessage, while there may be no consistent menu with undo or redo buttons, there is a lot of space to simply read and send messages.
Hypothetical Ideal Design	3	3	A hypothetical ideal design has a positive rating of 3 for both the operation indicators and interface simplicity. A user should be able to activate these operations easily via clear **signifiers** in order to reduce the **gulf of execution**. On top of that, this design should have a clean interface to reduce any strain and stress that may be induced onto the user.
Our Redesign	-1.5	2	Our redesign received a -1.5 for operation indicators as the feature of cross-platform communication is visually accessible to the user. Some forms of communication like Gmail that may be included in this messaging hub have obvious operations for undo and redo, which can help users who do not know about hidden gesture-activated operations. On the other hand, our redesign received a 2 for interface simplicity as it may not be as clean as iMessage, with multiple messaging apps combined into one hub.

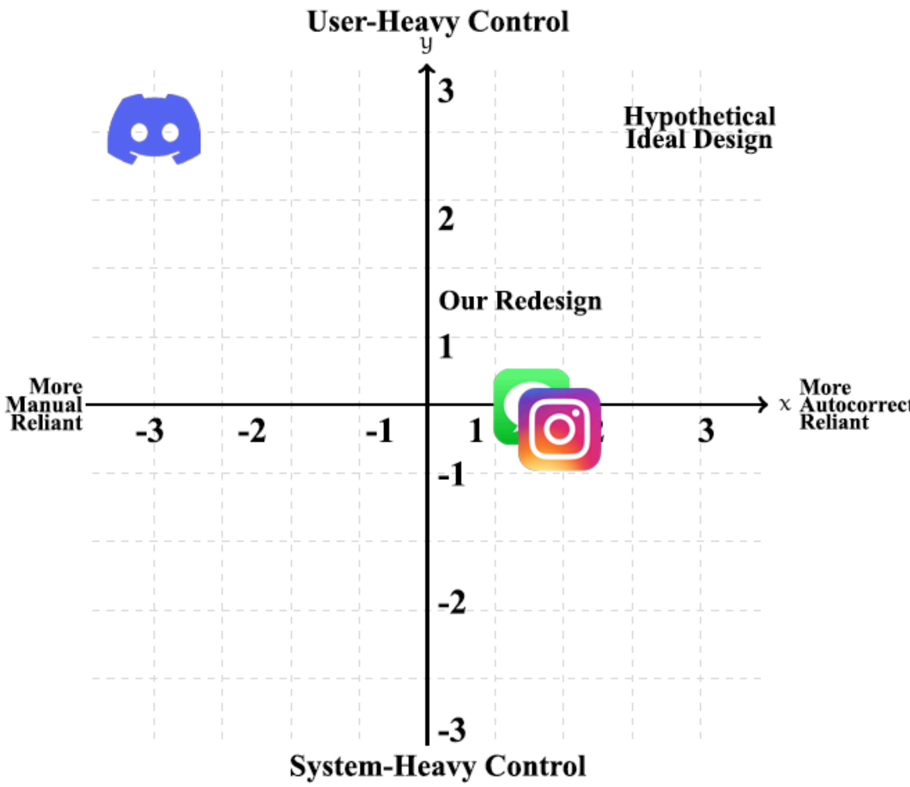

Figure 11. Our second design space chart that targets the **tradeoffs** between having autocorrect reliant vs manual reliant texting, this coming at a cost of the user's message accuracy and intent.

Chat Application	**Autocorrect Reliance (X-Axis)**	**User Control (Y-Axis)**	**Explanation**
iMessage	1.5	0	iMessage received a 1.5 for autocorrect reliance because by default, many users have autocorrect enabled, especially users of the older demographic that may not know how to disable it. So on average, iMessage can be deemed quite autocorrect reliant. On the other hand, iMessage has a 0 for user control, while this doesn't necessarily mean it has no user control at all, it is just in the middle between user-heavy and system-heavy control as seen in the chart. iMessage gives you the option to disable or enable autocorrect, choose between using the autocorrected word or not, but these features might not be immediately obvious to the user.
Discord	-3	3	Discord received a -3 for autocorrect reliance

			because it is a primarily PC application. On average, many computers do not have autocorrect built in (besides Apple computers/MacBooks), so this requires a lot more manual work in terms of messaging. There is less autonomy on the system's end and more reliance on the user, thus it received a 3 for user control.
Instagram	1.5	0	Instagram's direct messaging design is quite similar to iMessage. So like iMessage, it received a 1.5 for autocorrect reliance. As aforementioned, on average users have autocorrect enabled, thus this design being more autocorrect reliant. However, it is in the middle between user and system heavy control, thus receiving a 0. Users have the ability to disable or enable autocorrect, choose between using the autocorrected word or not, but these features might not be immediately obvious to the user.
Hypothetical Ideal Design	3	3	A hypothetical ideal design would have the most user-heavy control while also being autocorrect reliant. As contradictory as it may sound, autocorrect provides efficiency in typing, especially for users of the older demographic or users who have trouble spelling certain words. This is where autocorrect reliance would come into play and be quite helpful. On the other hand, our interviews have showed counts of overcorrection that ends up flawing the user's original intent in messaging which is why it should also be user-heavy controlled, so that users' authenticity and input is still valued when messaging, autocorrect just being a tool to help with typing.
Our Redesign	1	1.5	Our redesign received a 1 for autocorrect reliance, which is not too far off from iMessage and Instagram. This is because as aforementioned, it depends on the user's preferences that were set in regards to autocorrect. Our redesign is more catered towards solving cross platform communication. Our redesign received a 1.5 in user control because the user is no longer limited by the application when messaging other folks. The idea of a messaging hub breaks this barrier that users have to stick to one messaging

			application when messaging certain people, it opens them to a wider selection of applications.

Lo-fi redesign:

A potential solution to double texting across multiple platforms–and the conflicting **mental models** they propose–might come in the form of a cross-platform message hub. This will give users the ability to check one application for all their communication needs. Users will be able to pool selected apps and also isolate the original apps if they choose. A cornerstone of the new app, dubbed “OneChat”, will be customization to user preferences and use of smart text replacement. There will be a main application icon, for familiarity, located on the user desktop, home screen or start menu which users can access. This will provide the array of digital **affordances** that come from the classic desktop icon format. A single app means easy searching via a single search bar, which eliminates the **knowledge-based mistake** of not knowing where the search feature is on iMessage or other messaging applications. It also eliminates the issue of texting the same person on multiple platforms, synthesizing communication into a single hub. Furthermore we will provide the user with an overlay that follows them through the different apps when they leave OneChat hub. See Figure 12

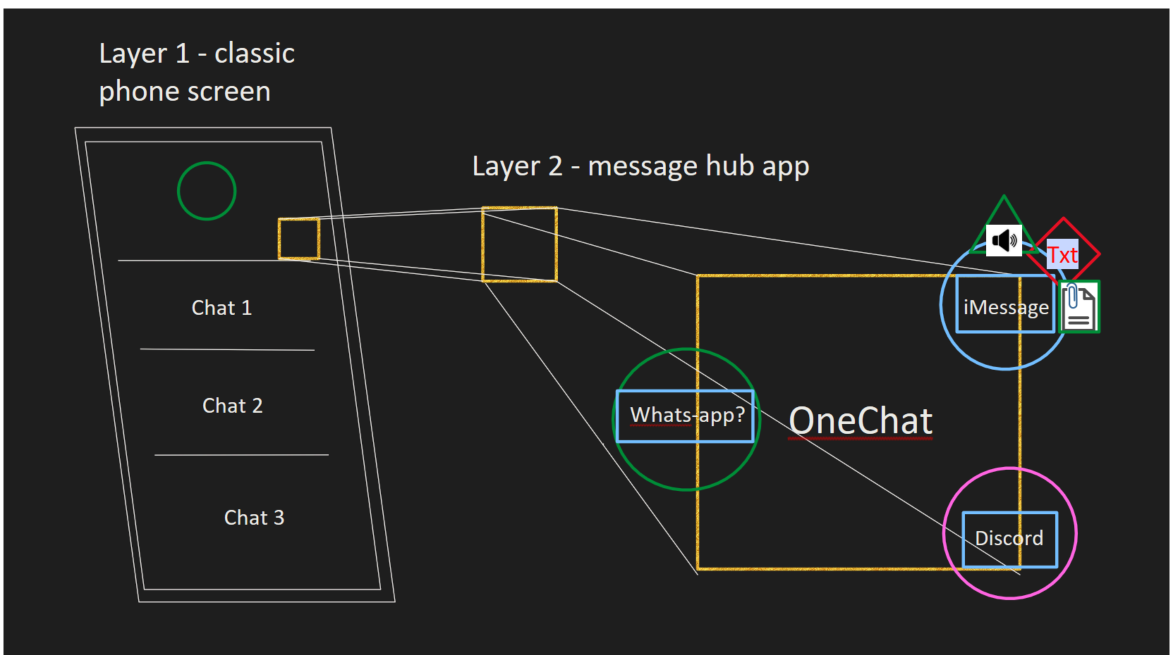

Figure 12. A sketch design of the widget like overlay, which will accompany the user into their original application programs. As a prototype it does a lot to explain what we are talking about.

An application with similar mobility can be found in the asus portfolio of tools, with the ScreenExpert3. See *Figure 13* for the ScreenExpert3 imagery. The SX3 app is available as an overlay onto all programs.

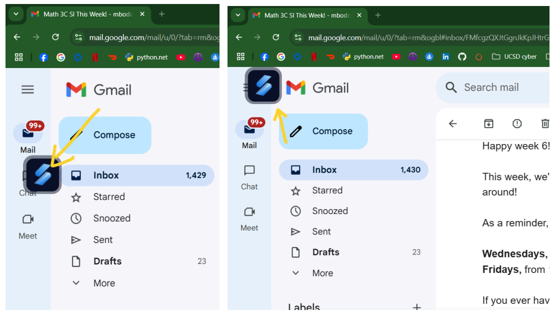

Figure 13a. and ***13b.*** The ScreenExpert3 icon floating above the UCSD gmail.

OneChat desktop hub can help reduce typing **slips**, such as the autocorrect issues we found in our first task. For example, the platform could leverage AI tools to adapt to users' personal texting styles and languages so that users do not experience unwanted autocorrect suggestions on words or phrases. Additionally, the platform could adapt to user input by asking for feedback before making autocorrect changes, other than the **slips** that are clearly spelling errors. Our new platform would also have a synthesized and clear edit menu. Simply tap next to your sentence to pull up the hover menu that has an undo and redo button, avoiding multiple menus or hidden features that **afford** undoing but are not **discoverable** (Figure 14). This helps avoid the **knowledge-based mistake** users were prone to making with the confusing three-finger tap and shake to undo features.

Figure 14: A simple UI hover menu from [Uxcel] synthesizes undo and redo in a single menu.

In virtual meeting interviews where users' first choices were not iMessage, 6 of 16 users were unable to complete the attachments task because their preferred platform did not provide the ease of sending attachments. But if the user had OneChat to guide them through the communication tools, they would easily identify key **signifiers** to send a simple attachment. Adding a clear symbol near the send button, such as those in Figures 15a and b, would **signify** that the app **affords** uploading and sending attachments. The modes of cross platform support will lower instances of **knowledge-based mistakes**.

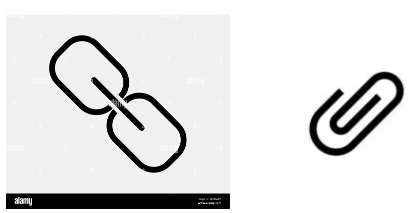

Figure 15a, 15b: Universal symbols for adding attachments or files.

In our redesign efforts we have identified a set of top **errors**: **knowledge errors** and **mode errors** pertaining to the inability to send attachments, **errors** of confusion with juggling conversations, and **errors** of sending a message to the wrong person. The inability to send an attachment is significant and is a crippling limitation to everyday communication, where a picture is worth 1000 words or a data sheet is the topic. This **error** is significant to artists and data scientists as well as many other groups.

Figure 16. Shows a designer able to send 5GB of work over to a group member. Illustrating the importance of this mode of metadata link.

As we have suggested earlier, the overlay tool will follow the user through apps providing support to users where the active app is missing this feature. We also encountered many people who confess that juggling communication can be difficult and can lead to the critical **error** of sending a message to the wrong person. If a user is texting someone and also emailing, while linked in discord they can lose track of what is going on, leading to **slips**. When we are able to pool the communication into the OneChat app they will be able to see the communication and have the option to consolidate into one messaging conversation, stopping the juggling. If the juggling has ever been so much that the user has messaged the wrong person, it is a critical **error** that can have real and serious consequences. This **error** is significant to people who live a fast paced life. We will avoid this **error** by having large text for the recipient name and a profile photograph which we will prompt the user to fill out early and with a recurring reminder; these are universal and easy to detect **signifiers**.

Redesign tradeoffs:

There are a variety of **tradeoffs** when switching over to a new system. It may take time for some users to become accustomed to the new interface. If we offer too much customization the user may begin to get confused by the variability of the user experience. The app will also obviously take up more space on the phone and computer than the single apps will. Furthermore, the app will also use more computing power, which may require a stronger device; creating a **tradeoff** of additional expense and possible lag. The inner app overlay may also visually block some small zone of the screen. Overall the **tradeoffs** are well worth the benefits. In short, there may be a steep learning curve but in the long term users will save time and efficiency in communication. As users begin to understand the new interface they will appreciate the cross platform availability. There will be little chance to miss a message as alerts will populate across platforms. We also have to consider the **tradeoffs** of having a simpler interface. Some may say this is good, but obviously the user will lose special features and have less customization, with less features overall.

We also have the **tradeoffs** from the design space to consider: Cleaner interface vs cluttered interface, more obvious operations vs more hidden operations, heavy user control vs heavy system control, more autocorrect reliant vs more manual reliant. Each zone will play a part in the designs of the new user experience. A clean interface may sacrifice the full tools. A user experience with more hidden operations may be liked by some but others may want to see things like the processing bar. This can be customized. A design with heavy user control may be tedious whereas a system control may seem out of control. We will offer different ways for the user to control the tools. An app with more autocorrect may lead the population to be unable to spell with a stumble event if it is wrong. There are countless **tradeoffs** to consider.

Extra credit link!
https://docs.google.com/document/d/1H8ahC-Es5U-pkv77UgIuSNWBxJw8EyGtJFe1K4LvCTE/edit?usp=sharing